

Robert (Rob) J. Spenceley

Spenceley@wisc.edu | (720) 388-0242 | [My Website](#) | [Google Scholar](#) | [GitHub](#) | [LinkedIn](#)

SUMMARY

Human factors PhD student studying how humans interact with AI and embodied automation in high-risk and dynamic teaming environments (healthcare, search and rescue, emergency operations). This involves refining how information is presented, the form factor of AI systems, and the design of automation that enhances collaboration among human team members.

Keywords: Human-AI Teaming, Human-Robot Interaction, Human Factors, Neuroergonomics.

EDUCATION

University of Wisconsin – Madison (UW-Madison) Jan. 2025 – Present

Doctor of Philosophy – Industrial and Systems Engineering (Human Factors)

Advisor: Dr. Ranjana K. Mehta

Expected Graduation Date – Fall 2028

Colorado State University – Fort Collins (CSU) Aug. 2021 – Dec. 2024

Bachelor of Science – Psychology (Concentration: Mind, Brain, & Behavior)

Minor: Business Administration

RESEARCH EXPERIENCE

Graduate Research Assistant – Neuroergonomics Lab Jan. 2025 – Present

Department of Engineering – Industrial and Systems Engineering

University of Wisconsin-Madison

Principal Investigator: Dr. Ranjana Mehta

Projects: Physical Human-Robot Interaction (pHRI), Trust in Multi-Human Robot Teams

- Conduct experiments using a breadth of bioinstrumentation (EEG, fNIRS, eye-tracking, AMTI Force Plate, EMG, & ECG) and a UR10e robotic arm.
- Streamline data collection using methods such as Lab Streaming Layer, Python, & MATLAB.
- Analyze EEG data collected from a virtual reality disaster robotics experiment and pHRI, using power spectrum density, corticomuscular coupling, and hyperscanning analyses.

National Science Foundation (NSF) Research Traineeship (NRT+) Trainee Jan. 2025 – Present

Department of Computer Science - Integrating Robotics into the Future of Work (INTEGRATE)

University of Wisconsin-Madison

Principal Investigator: Dr. Bilge Mutlu

Project: Comfort Survey (CHIRP)

- Lead a team of RAs to develop virtual reality experiments.
- Develop a user research survey to quantify human comfort when working with robots in occupational settings.
- Collect and analyze survey data using methods like exploratory factor analysis, confirmatory factor analysis, and the Delphi Method.

Lab Manager – Wickens Lab Aug. 2024 – Dec. 2024

Department of Psychology – Cognitive Psychology

Colorado State University

Projects: FIPRED, Augmented Reality Cueing in Visual Search, Germane Load.

- Developed FIPRED, a study of predictive automated displays for wildfire fighting management, and collected data for it and for graduate student studies. Despite finding predictive displays to be beneficial to participants' ability to predict wildfire trajectory, we find evidence that the displays create automation bias when wrong, motivating a need to calibrate optimal reliance before implementation.

- Collected over 150 hours of data using the HoloLens 2, Meta Quest 3, and wide-angle displays.
- Oversaw data collection schedules and maintained clear communication with participants throughout the study process.

Lab Manager – Davalos Lab

Jul. 2023 – Dec. 2024

Department of Psychology – Cognitive Neuroscience
Colorado State University

Project: Neural Correlates of Prospective Memory

- Trained new undergraduate research assistants/graduate students in EEG, oversaw the creation of data collection schedules, ensured the proper maintenance of lab equipment, & assisted with other undergraduate projects using the lab's data.
- Conducted over 300 hours of EEG data collection, building on prior experience as a research assistant.

LEADERSHIP

Human Factors & Ergonomics Society – UW-Madison Chapter

Chapter President

Spring 2026 – Present

Vice President of Events

Fall 2025 – Spring 2026

PUBLICATIONS

Spenceley, R. J., Yadav, A., & Mehta, R. K. (Accepted). An Integrated Multimodal Framework for Measuring Physical Team Processes During State Emergency Operations Center Activation. *Sage Journals In Proceedings of the Human Factors and Ergonomics Society Annual Meeting*. Sage CA: Los Angeles, CA: SAGE Publications

Spenceley, R. J., & Mehta, R. K. (2026). When robots reshape teams: Neurodynamic insights into taskwork and teamwork in search and rescue. *Frontiers in Neuroergonomics*, 7.
<https://doi.org/10.3389/fnrgo.2026.1771753>

Spenceley, R., Wickens, C., & De Toledo Piza, B. (2025). FIPRED: A study of predictive automated displays for wildfire fighting management. *Sage Journals In Proceedings of the Human Factors and Ergonomics Society Annual Meeting* (pp. 1-7). Sage CA: Los Angeles, CA: SAGE Publications.
<https://doi.org/10.1177/10711813251364799>

Wickens, C., & Spenceley R. (In Press). The trajectory prediction challenge (TPC): An application of cognitive engineering. In Khalid, H. M. (Eds.), *Helander's compendium of human factors and ergonomics*.

PRESENTATIONS

Spenceley, R. J., Yadav, A., & Mehta, R. K. (2026). *An Integrated Multimodal Framework for Measuring Physical Team Processes During State Emergency Operations Center Activation*. ASPIRE, Reno, Nevada. – **Lecture**

Spenceley, R. J., & Mehta, R. K. (2026). *When robots reshape teams: Neurodynamic insights into taskwork and teamwork in search and rescue*. Neuroergonomics Conference, Boston, Massachusetts. – **Lecture**

Spenceley, R. J., & Mehta, R. K. (2026). *Are You Ready for This? An Evaluation of Motor-Readiness Cortical Potentials While Co-Lifting with an Unreliable Robot Partner*. Neuroergonomics Conference, Boston, Massachusetts; HFES ASPIRE, Reno, Nevada. – **Poster**

Spenceley, R., Wickens, C., & De Toledo Piza, B. (2025). *FIPRED: A study of predictive automated displays for wildfire fighting management*. Human Factors and Ergonomics Society – ASPIRE, Chicago, Illinois. – **Poster**

Spenceley, R., Yadav, A., & Mehta, R. (2025). *EEG theta power varies across roles in mixed human-robot teams*. Human Factors and Ergonomics Society – ASPIRE, Chicago, Illinois. – **Poster**

Spenceley, R., Rice, M., & Davalos, D. (2024). *Components of executive functioning involved in prospective memory*. Rocky Mountain Psychology Association Conference, Denver, Colorado. – **Poster**

REVIEWER

TEACHING EXPERIENCE

Instructor PSY 292C – Controversial Issues in Psychology	Fall 2024
Teaching Assistant PSY 252 – Mind, Brain, & Behavior	Spring & Summer 2022, Summer 2024
Teaching Assistant PSY 250 – Research Design & Analysis I	Fall 2022

HONORS & AWARDS

College of Natural Science Dean's List - <i>CSU</i>	2022, 2023, & 2024
Chancellor's Opportunity Award - <i>UW-Madison</i>	2024

RELEVANT SKILLS

Research Techniques: EEG/fNIRS cap placement/fitting (10-20), EMG, ECG, human motion tracking (Xsens), eye-tracking (Tobii), HoloLens 2, Meta Quest 3, & computer vision.
Software: R, MATLAB (EEGLAB/ERPLAB), BrainVision, Homer3, Aurora, Noraxon, Tobii, Lab Streaming Layer (LSL), Curry7, PsychoPy, Python, Qualtrics, Shimmer, Prolific.
Certifications: Behavioral/social CITI Training, Good Clinical Practice CITI Training, & Delta Program Mentorship Certificate.

MEMBERSHIPS

Human Factors and Ergonomics Society (HFES) – Student Affiliate	2025 – <i>Present</i>
Human AI Teaming (HART) Technical Group (HFES)	2026 – <i>Present</i>

APPEARS IN

UW College of Engineering Editorial – [How do robots impact search-and-rescue teamwork? Engineers look at human brain activity to find out](#) – 4/13/26
[RIT AWARE AI Newsletter](#) – December 2025